

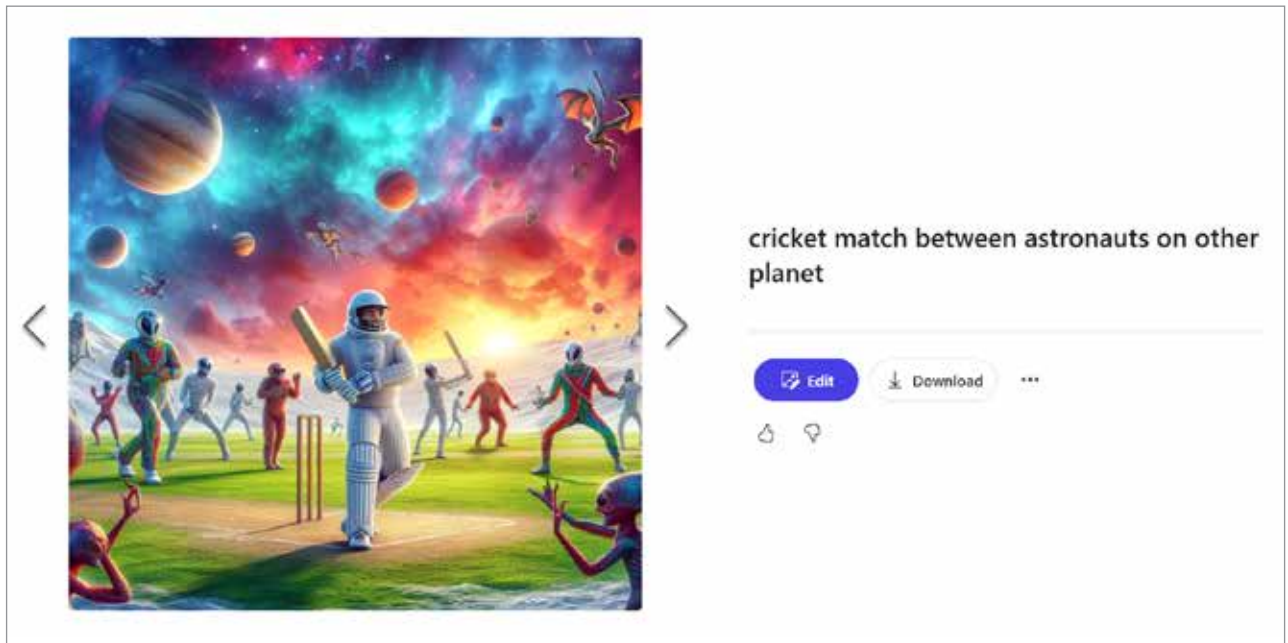


Figure 1: Generation of a hypothetical scenario using Microsoft Designer

environments into separate containers.

This is especially helpful for workflows involving generative AI, which frequently use sophisticated libraries like Hugging Face Transformers, TensorFlow and PyTorch. By offering reproducible environments, which are essential for research using large language models (LLMs), image synthesis, or multimodal AI, Docker makes it easier for data scientists and developers to collaborate.

Docker is frequently used to expedite the creation and implementation of AI chatbots, image generators, synthetic data pipelines and real-time inference services, among other use cases. In order to enable scalable and flexible AI systems, it also supports microservices architectures, in which various generative components such as text creation, image rendering, or voice synthesis run in distinct containers.

In addition, Docker streamlines CI/CD pipelines for continuous training and deployment in research or production contexts, and enables the edge deployment of lightweight AI models in IoT or mobile applications. Docker enables practitioners of generative AI to create, test and deliver intelligent applications effectively and consistently.

Common use cases of Docker and generative AI are:

- Text generation with GPT models.
- Utilising FastAPI or Flask, packaging a GPT model (such as GPT-Neo, GPT-J, or LLaMA) and implementing it via a REST API.
- Stable diffusion for image generation.
- Making use of Dockerized environments with support for the GPU to carry out inference pipelines using models like Stable Diffusion.

Table 1: How generative AI is useful for businesses

GenAI activity	Application
Creativity amplification	Creativity and innovation
Implementation barriers	Job displacement
Product development	Customer service
Ethical concerns	Ethical frameworks and AI regulation
Ethical considerations	Improving decision making with predictive analytics
Automated content production	Automation
Quality control	Risk management
Generative AI techniques	Improved productivity
Marketing and advertising	Optimise product designs
Automation and efficiency	Bolster cybersecurity
Personalised customer experiences	Personalisation
Content creation and design	Mathematical optimisation

- Creating developer tools that, when containerised for CI/CD deployment, use LLMs to generate boilerplate code.
- The virtual assistant and chatbot containerise the